



Bullittsville Pump Station

CENTRAL BOONE COUNTY **SEWER PROJECT**

ALTERNATIVES EVALUATION AND SELECTION OF B.2

Prepared by
Sanitation District No. 1
of Northern Kentucky

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CENTRAL BOONE COUNTY CONVEYANCE

Report on the Selection of Alternative B.2

Sanitation District No. 1 of Northern Kentucky

Prepared for the SD1 Board of Directors

EXECUTIVE SUMMARY

SD1 selected Alternative B.2 for the Central Boone County conveyance project to address existing wastewater system needs, including the elimination of approximately 1.2 million gallons of sanitary sewer overflows into Woolper Creek in a typical year, while establishing a more efficient and flexible long-term conveyance system. Although the alternatives had relatively comparable planning-level costs, B.2 provided greater hydraulic reach, increased future gravity-service potential, and reduced long-term reliance on pump stations, equalization facilities, and other mechanical infrastructure which are costly to maintain and operate.

The construction, environmental, property, and easement risks identified during planning were carefully evaluated and have been effectively managed. All required environmental permits have been obtained, necessary easements have been acquired, potential cost overruns have not materialized, and the project is significantly under budget.



1. Purpose of This Report

This report summarizes the history, alternatives and long-term infrastructure considerations that informed SD1's selection of Alternative B.2 for the Central Boone County conveyance project. It is intended to provide the SD1 Board, local officials and members of the community with a clear explanation of the decision and the tradeoffs considered during planning.

2. Background: Lessons from the Hebron KY 237 Corridor

SD1's planning for Central Boone County was influenced by its experience with the wastewater systems along the Hebron KY 237 corridor. The KY 237 corridor relies heavily on interconnected pump stations and force mains which were added incrementally as development occurred. Over time, the resulting network of pump stations created bottlenecks and additional infrastructure that had to be operated, maintained, upgraded and eventually replaced.

In 2013, after the funding needed to upgrade these systems was unavailable, the Boone County Fiscal Court formed the Hebron Sewer Study Committee to address the sewer challenges along the Hebron KY 237 corridor. In its March 2014 report to the Fiscal Court, the committee identified the need for a more comprehensive approach to sewer planning, which included a new gravity sewer system and major transmission line directing wastewater toward the Western Regional Treatment Plant. At that time, SD1's Consent Decree obligations and financial constraints made that approach infeasible.

That experience became an important planning consideration for Central Boone County. Rather than repeat a pattern of adding individual pump stations as needs arose, planners considered the most efficient and effective way for the wastewater system to function over the long term.

3. Planning for Central Boone County

In the early 2010s, financial constraints plagued SD1. Operating under a federally mandated consent decree with the U.S. Environmental Protection Agency (EPA), SD1 developed overflow remediation plans focused on meeting the decree's requirements at the lowest cost. To examine ways to eliminate sanitary sewer overflows into Woolper Creek, SD1's consultant, Hazen, completed the "Bullittsville Pump Station Flow Routing Business Case Evaluation" in March of 2017 (the "Hazen report"). The Hazen report identified three engineering approaches for accomplishing that objective with similar costs: alternatives A.3, B.2 and B.3.

At the time of the Hazen report, SD1's financial outlook remained unchanged and immediate improvements to the Bullittsville Pump Station were necessary to meet consent decree requirements. With these upgrades already required, Option B.3 emerged as the most viable alternative for redirecting flows to Western Regional because it would leverage the planned investment at Bullittsville. However, it was clear under the circumstances that B.3 would have to wait a decade or two for funding. The plans to upgrade Bullittsville Pump Station and the remaining A3 option were moved forward with the remainder of the B3 option put on hold until the consent decree obligations were satisfied.

Beginning in 2017, SD1 pursued several major initiatives to strengthen its financial position and expand its ability to serve the community. Primarily among them was the renegotiation of its consent decree in 2019. This marked a significant turning point, providing SD1 with greater flexibility to meet its regulatory obligations while more effectively fulfilling its broader community mission. Consistent with the planning



information available at the time, SD1's Updated Watershed Plan incorporated the A.3 approach identified in the Hazen Report.

As SD1's financial position improved, the utility asked each county to identify its highest-priority sanitary sewer needs for consideration in long-term financial planning. In a February 9, 2021, response, Boone County identified limitations associated with the Bullittsville and Taylorsport Pump Stations, force mains and gravity sewers as its top sanitary sewer priority. The county reiterated its support for a longer-term strategy to redirect wastewater toward the Western Regional treatment system, where available treatment capacity could be used and system constraints could be avoided.

Armed with more financial flexibility and clearer direction on priorities, SD1 re-evaluated the Hazen options to improve system reliability and overflow performance, reduce long-term asset-management requirements and address Boone County's identified sewer priority with a single long-term investment that would benefit ratepayers and support the economic vitality of the region.

The development of the Central Boone County conveyance project occurred over several years and involved multiple planning, engineering, funding and public-process milestones. A chronology of the key events leading to the selection and implementation of Alternative B.2 is provided in Appendix 4.

4. Alternatives Evaluation

Regardless of which alternative was ultimately selected, the primary purpose of the project was the same: to address the existing wastewater system needs in Central Boone County, including the elimination of approximately 1.2 million gallons of sanitary sewer overflow into Woolper Creek in a typical year.

The re-negotiation of the consent decree allowed the initial improvements to Bullittsville pump station to be delayed. Since no large investments had been made to Bullittsville pump station, SD1 still had three viable options. Therefore, the question before SD1 was which alternative would provide the most appropriate long-term solution when considering engineering, regulatory requirements, constructability, lifecycle costs, environmental impacts, system reliability and long-term community needs.

Alternative A.3

Alternative A.3 offered a more straightforward construction approach but would continue relying on the existing Bullittsville and Taylorsport Pump Station conveyance system. It would require additional equalization facilities at both Bullittsville and Taylorsport, along with property acquisition for those facilities. As a result, SD1 would continue operating and maintaining multiple pumping and equalization facilities over the life of the system.

Alternative B.2

Alternative B.2 uses a lower-elevation gravity alignment and the Central Boone County Pump Station. It eliminates the Bullittsville Pump Station from the long-term conveyance strategy and avoids the Bullittsville equalization facilities required under B.3, as well as the new equalization facilities at Bullittsville and Taylorsport required under A.3. Its lower elevation provides greater hydraulic reach, but the alignment presents greater construction, terrain, environmental and property/easement challenges than the other alternatives.



Alternative B.3

Alternative B.3 would route wastewater toward the Central Boone County Pump Station but would require an upgrade to the Bullittsville Pump Station, a new force main and an above-ground equalization tank at Bullittsville before transitioning to gravity conveyance. Its gravity sewer would begin at a higher elevation than B.2, providing less hydraulic reach.

The planning alignments for Alternatives B.2 and B.3 are illustrated in Appendix 2.

5. Why B.2 and B.3 Differ

The central distinction between B.2 and B.3 is not simply the route of the sewer. It is the type of wastewater system each alternative establishes for future decades.

B.3 begins with a pumped system before transitioning to gravity conveyance, while B.2 establishes gravity conveyance at a lower elevation. That difference matters because the lower the gravity sewer sits within the watershed, the greater the area that can potentially drain to it without additional pumping.

As a result, the B.2 configuration provides greater hydraulic flexibility while reducing the amount of pumping and equalization infrastructure SD1 would ultimately own, operate, maintain and replace in perpetuity. The maps in Appendix 3 illustrate the future areas that could potentially be served by gravity sewers under Alternatives B.2 and B.3. The shaded areas represent areas that could reach the gravity sewer without requiring additional pump stations, thereby mitigating the long-term issues experienced through the development of the Hebron KY 237 corridor.

A side-by-side comparison of the principal planning differences and capital-cost estimates for Alternatives B.2 and B.3 is provided in Appendix 1.

6. Why Gravity Sewer Is Preferred

Where topography allows, gravity sewer is preferred because wastewater moves through the system without continuous mechanical pumping. This reduces dependence on electricity and mechanical equipment and can lower long-term operational complexity. The comparison below summarizes the basic differences between gravity conveyance and a pump station/force main system.

Metric	Gravity Sewer	Pump Station & Force Main
Energy Consumption	None (100% passive)	High (ongoing electrical costs)
Moving Parts	Zero	High (pumps, valves, float switches)
Equipment Lifespan	50-100+ years (pipes/manholes)	10-20 years (pump cores/electronics)
Risk of Failure	Low (mainly localized blockages)	Moderate to high (power outages, mechanical failure)
Odor & Septicity	Low (continuous, passive flow)	Higher (sewage sitting in wet wells can turn septic)



For SD1, the long-term simplicity of gravity conveyance was an important planning consideration. Pump stations and force mains remain essential where elevation makes gravity impractical but maximizing gravity conveyance where feasible reduces dependence on mechanical and electrical infrastructure over the life of the system.

7. Construction, Environmental and Property Considerations

The Hazen report identified that B.2 carries greater near-term construction, terrain, environmental and property/easement challenges and greater risk of cost overruns than B.3. The selection of B.2 therefore reflected a long-term infrastructure tradeoff: greater complexity during construction in exchange for a lower-elevation gravity system, greater hydraulic flexibility and reduced reliance on mechanical pumping infrastructure over the life of the system.

A significant component of the environmental work associated with B.2 involves crossing streams and other regulated waters along the gravity sewer alignment. These crossings require more than simply installing a sewer across a waterway. SD1 must obtain the applicable federal and state environmental authorizations, minimize impacts to streams and wetlands, establish limits of disturbance, and implement appropriate measures during construction to protect the waterways. Following construction, temporarily disturbed areas must be restored, including restoring contours and stabilizing and re-establishing appropriate native vegetation where required.

For B.2, this means that the environmental work is not limited to obtaining a permit before construction. SD1 must manage the environmental requirements throughout construction and complete the required restoration and mitigation following the work, including restoring areas temporarily disturbed for construction access and stream crossings. These requirements are part of the project's construction obligations and are monitored as the project progresses.

On February 10, 2022, SD1 staff invited Gayle Pille to a public board working group meeting to address her concerns regarding environmental impacts. SD1's Environmental Scientist presented to the Board and the guests in attendance the strategies SD1 utilizes to mitigate the environmental impact the new infrastructure will have on the watershed. SD1's Environmental Compliance Manager then provided information to the Board regarding the water quality in Woolper Creek. She explained that SD1 sees opportunities to address those impairments through the infrastructure project using successful asset management approaches.

On March 1, 2022, SD1 advertised and conducted a public meeting to get community input on the project. The meeting was well attended, and no significant hurdles were identified.

The risks and property challenges associated with B.2 were carefully evaluated during the planning process, and staff determined that they could be effectively mitigated through appropriate project management and risk controls. That assessment has been validated by the project's performance. The risks and disadvantages initially identified with the B.2 alternative were successfully resolved. The evaluation identified no conservation areas, and all required environmental permits have been obtained from the U.S. Army Corps of Engineers and the Kentucky Division of Water. The environmental requirements associated with the permits have been incorporated into project design and construction, including the required restoration and mitigation measures.



The necessary easements were obtained through negotiation or were awarded by court order. Additionally, the potential cost overruns did not occur. Through disciplined project management, effective risk mitigation, design changes, favorable bids and successful execution, the Central Boone County conveyance project with the B.2 route is projected to achieve a final cost of approximately \$39.5 million. And because Boone Fiscal Court directed the use of approximately \$10.6 million in ARPA funds towards the project, SD1's net cost will be under \$30 million. The cost savings and the project's performance support staff's original assessment that the risks associated with B.2, identified in the Hazen report, could be effectively managed while preserving the long-term benefits of the B.2 alternative for the ratepayers.

The completion of the sewer project would not prevent Ms. Pille from pursuing conservation of her property. Following completion of the project, Ms. Pille may still choose to donate the property to a land conservancy or similar organization for designation and long-term management as a nature preserve, subject to the terms and requirements of the receiving organization and any applicable easements. SD1 is willing to assist in facilitating that process once the sewer project and associated restoration activities are complete.

8. Basis for SD1's Selection

SD1 selected B.2 after weighing near-term construction impacts against long-term system performance and asset-management considerations. The principal factors included:

- a lower-elevation gravity configuration that provides greater hydraulic reach and flexibility;
- elimination of the Bullittsville Pump Station and avoidance of additional Bullittsville equalization infrastructure;
- reduced long-term reliance on pumping infrastructure and the associated operation, maintenance and asset-management obligations; and
- ability to provide gravity sewer service to currently unserved areas as future needs arise.

BOTTOM LINE

B.2 was ultimately selected as the preferred alternative because it addresses SD1's current wastewater needs while establishing a long-term system designed to maximize gravity conveyance, minimize reliance on additional pump stations and other mechanical assets and was the only alternative that eliminated the Bullittsville Pump Station. The selection intentionally accepted greater construction complexity in the near term in exchange for reduced long-term operational complexity, greater system resiliency, and lower lifecycle costs.

The risks identified for the B.2 alternative in the Hazen report were carefully evaluated and have been effectively managed throughout project development and construction. The contracts for all phases have been awarded, and project performance has exceeded expectations. Based on its ability to meet current system needs while providing long-term operational and financial benefits, B.2 provides the greatest long-term value for SD1 ratepayers.



Appendix 1: B.2 and B.3 Evaluation Comparison

The following table summarizes the principal planning differences and capital-cost estimates for Alternatives B.2 and B.3.

B.2 and B.3 Evaluation Comparison

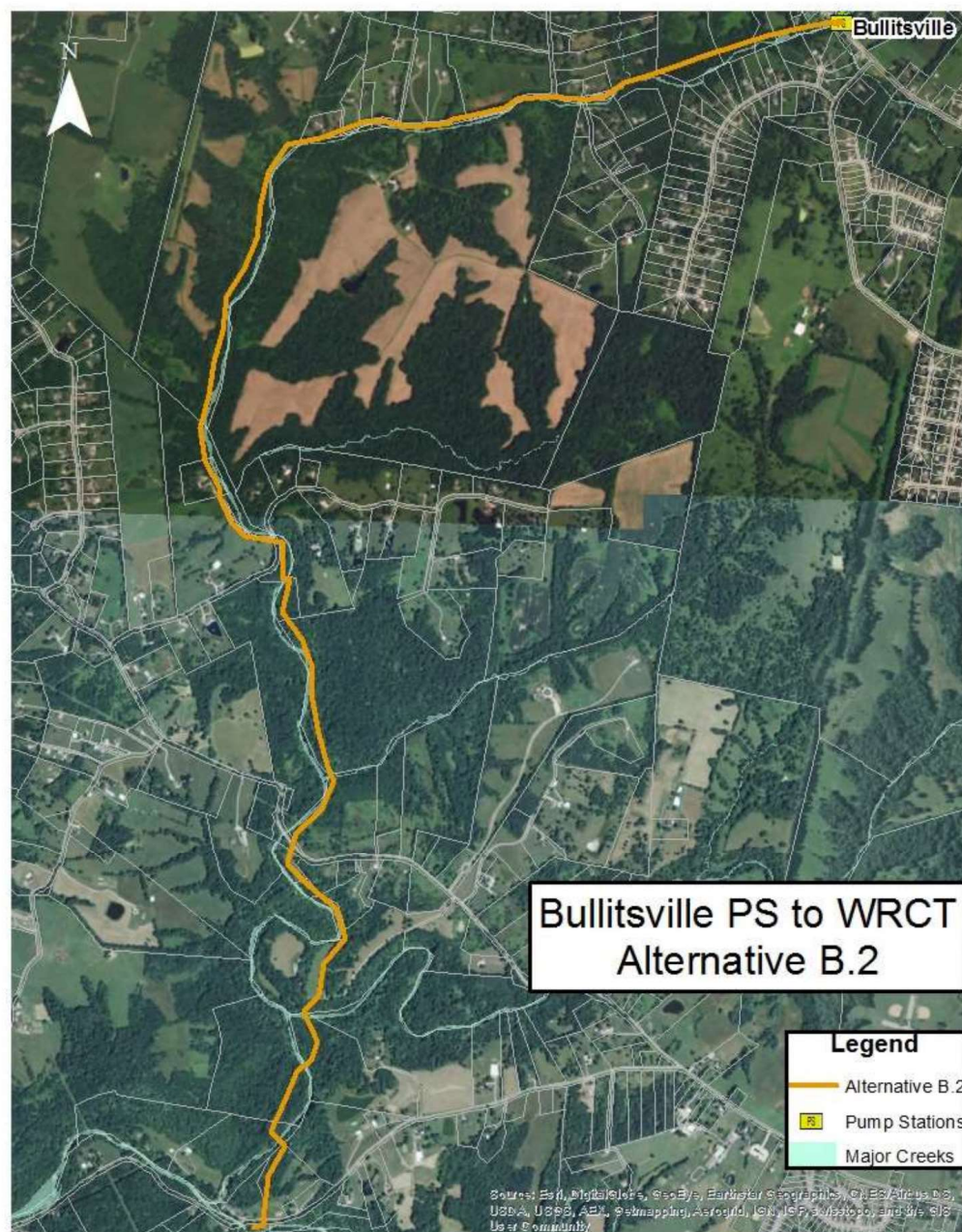
Evaluation Area	B.2	B.3
Overall Conveyance Approach	Provides a predominantly gravity-based conveyance system from the Bullittsville area to the Central Boone County Pump Station.	Upgrades Bullittsville Pump Station and adds a new force main before transitioning to gravity conveyance to the Central Boone County Pump Station.
Pump Station & EQ Requirements	Eliminates the need for Bullittsville Pump Station and associated EQ facilities as part of the long-term conveyance strategy.	Requires Bullittsville Pump Station upgrades and an above-ground EQ tank but avoids the need for EQ at Taylorsport.
Hydraulic / Service-Area Flexibility	Provides the greatest hydraulic reach and future service-area flexibility due to its lower-elevation gravity alignment.	Provides future service opportunities but is constrained by the higher elevation at which the gravity system begins, resulting in less hydraulic reach than B.2.
Property Requirements	Requires property/easement acquisition along the new gravity alignment but avoids property acquisition for the EQ facilities required under A.3 and B.3.	Requires property/easement acquisition along the alignment and property for the Bullittsville EQ facility.
Construction / Environmental Considerations	More complex construction with greater relative environmental, terrain, and easement challenges. Environmental permitting requirements have been addressed, and all required environmental permits have been obtained.	Generally, there is less construction and environmental complexity than B.2 due in part to the higher alignment.
Long-Term Operations	Reduces long-term reliance on pumping infrastructure by utilizing gravity conveyance and eliminating Bullittsville Pump Station.	Requires continued operation of Bullittsville Pump Station, its new force main, and the Bullittsville EQ facility as well as greater reliance on additional pump stations in future service areas.
Capital Cost	\$73,200,000	\$75,700,000
Overall Tradeoff	Greater near-term construction complexity in exchange for the greatest long-term hydraulic flexibility, reduced pumping infrastructure, and future service potential. Overall best value for ratepayers.	Less construction complexity than B.2, but greater long-term reliance on pumping and higher-elevation gravity system that limits future service potential and requires higher long-term costs for ratepayers.

Note: Cost figures are planning estimates.

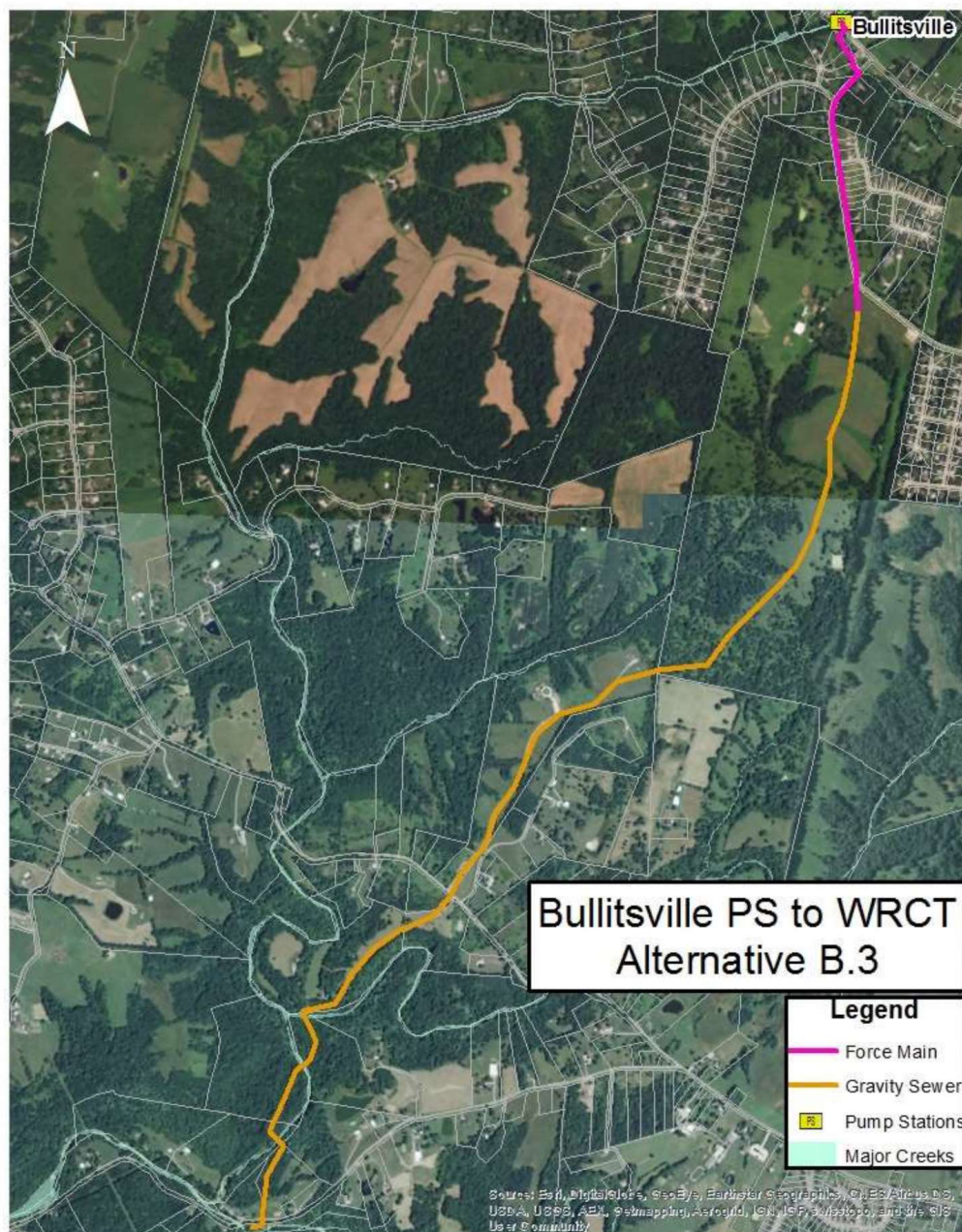
Appendix 2: Alternative B.2 and B.3 Conveyance Alignments

The following maps illustrate the planning alignments for Alternatives B.2 and B.3 from the Bullittsville area toward the Western Regional conveyance system.

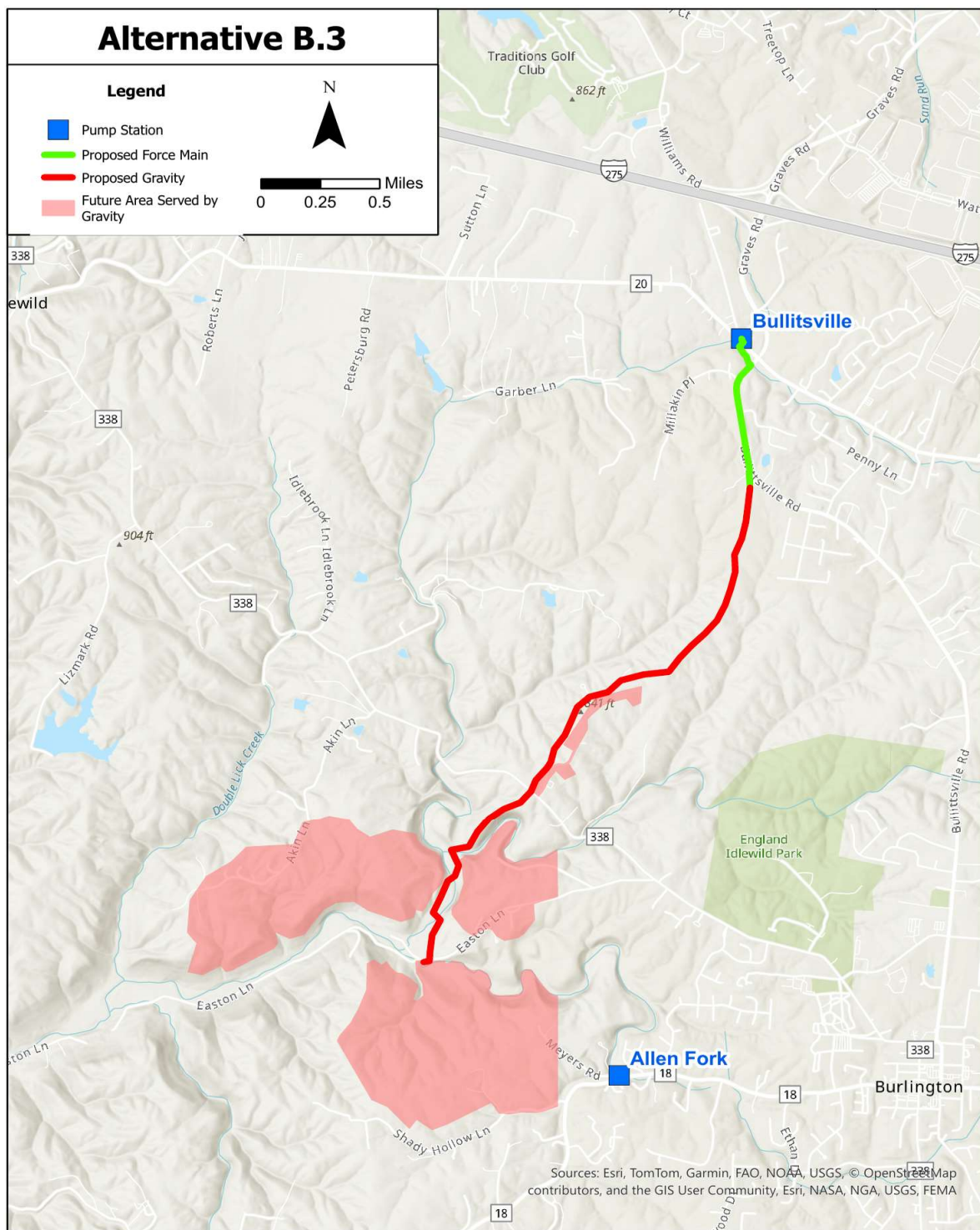
Alternative B.2 Conveyance Alignment



Alternative B.3 Conveyance Alignment



Alternative B.3 Gravity Service Area



Appendix 4: Project Timeline

The following timeline summarizes key milestones that informed the selection and implementation of Alternative B.2.

Date	Milestone	Significance to B.2 Selection
April 18, 2007	SD1 entered the Consent Decree	Established the regulatory and financial context under which major system improvements were evaluated.
September 17, 2013	Boone County Fiscal Court appointed a Hebron Sewer Study Committee to review sewer capacity constraints	A focused examination of wastewater capacity and conveyance challenges began in the Hebron KY 237 corridor.
March 11, 2014	Hebron Sewer Study presented to Boone County Fiscal Court	Identified limitations associated with interconnected pump stations and force mains and considered a more comprehensive gravity-based conveyance approach toward Western Regional. At the time, Consent Decree obligations and financial constraints made that approach infeasible.
March 14, 2014	SD1 submitted its watershed plan	Continued development of system-wide solutions to address sanitary sewer overflow and capacity needs.
February 5, 2016	SD1 issued RFP for a long-term solution to route Bullittsville flows toward the Western Regional Conveyance Tunnel	Initiated a formal engineering evaluation of alternatives to address the Bullittsville conveyance problem.
March 1, 2017	Hazen completed the Bullittsville Pump Station Flow Routing Business Case Evaluation	Evaluated A.3, B.2 and B.3 as viable, cost-effective alternatives. Planning-level costs were relatively comparable, so cost alone did not determine the outcome. This was an alternatives evaluation—not the final selection of B.2.
March 7, 2019	SD1 renegotiated the Consent Decree	Improved the regulatory and financial context for advancing long-term conveyance improvements.
February 9, 2021	Boone County identified Bullittsville/Taylorsport system constraints as its top sanitary sewer priority	Reaffirmed the need for a long-term solution and supported redirecting flows toward the Western Regional system.
March 24, 2021	SD1 Planning developed the Western Boone County Conveyance Project Charter	Formalized the renewed effort to advance a comprehensive conveyance solution.
July 20, 2021	SD1 Board approved Tetra Tech design contract for gravity conveyance for Phases 3 and 4	Moved the project from planning toward detailed design and implementation of the gravity conveyance concept.
January 18, 2022	SD1 Board approved easement acquisition services	Advanced property acquisition necessary for the selected project alignment.
February 10, 2022	Board Working Group meeting	Board listened to and addressed concerns regarding environmental impacts.
March 1, 2022	Public meeting held	Provided community/stakeholder engagement as the project advanced.
June 15, 2022	Updated Watershed Plan approved	Updated plans for system-wide solutions to address sanitary sewer overflow and capacity needs.
2022–2025	Detailed design, environmental permitting, easement acquisition and project development	B.2's planning-level challenges were worked through. Required environmental permits were obtained, and necessary easements were acquired or otherwise resolved.
December 16, 2025	SD1 Board approved Phase 3 construction contract	Advanced implementation of the B.2 conveyance system through award of the Phase 3 construction contract.
July 21, 2026	SD1 Board approved Phase 4 construction contract	Advanced implementation of the B.2 conveyance system through award of the Phase 4 construction contract.